

Yissum Research Intelligence Report — Week 30, 2026

12 high-potential paper(s) · Weekly · Week 30, 2026 · Generated July 22, 2026

Hebrew University researchers are advancing groundbreaking innovations across several high-impact areas, from novel cell therapies and antimicrobial solutions to cutting-edge AI platforms. Key developments include an off-the-shelf stem cell therapy for neurological repair, advanced single-cell sequencing for precision diagnostics, and a new generation of AI tools accelerating drug discovery and personalized medicine. These technologies address significant unmet medical needs and offer scalable solutions for a range of global health challenges.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Tamir Ben-Hur** — Neuroscience, Immunology, Clinical, Synthetic Biology
- **Oren Ram** — Genomics, Diagnostics, Drug Discovery, Clinical
- **Oren Zimhony** — Medical Device, Materials, Clinical
- **Eylon Yavin** — Drug Discovery, Genomics, Synthetic Biology
- **Mattan Hurevich** — Drug Discovery, Diagnostics, Materials, Synthetic Biology
- **Haitham Amal** — Drug Discovery, Neuroscience, Proteomics, Genomics
- **Ayelet N Landau** — Diagnostics, Medical Device, Software/AI, Neuroscience
- **Mor Nitzan** — Drug Discovery, Software/AI, Genomics, Immunology
- **Ofer Mandelboim** — Immunology, Drug Discovery, Clinical
- **Haitham Amal** — Drug Discovery, Neuroscience, Proteomics, Genomics
- **Yuval Ramot** — Software/AI, Immunology, Clinical, Drug Discovery
- **Ora Schueler-Furman** — Drug Discovery, Software/AI, Proteomics

ALL BRANCHES — RESEARCH HIGHLIGHTS

1. Pro-repair properties of a human embryonic stem cell-derived astrocyte cell therapy in demyelinating disorders. 36/50

"Off-the-shelf stem cell therapy promises neural repair for demyelinating diseases like MS."

Main researcher: Tamir Ben-Hur · tamir@hadassah.org.il.

Faculty of Medicine, Hebrew University of Jerusalem, Jerusalem, Israel; The Department of Neurology, The Agnes-Ginges Center for Human Neurogenetics, Hadassah-Hebrew University Medical Center, Jerusalem, Israel. Electronic address: tamir@hadassah.org.il.

Published in Stem cell reports · 2026-06

WHY NOW

Addresses a critical unmet need for remyelination, offering a scalable solution in a growing market for neurodegenerative diseases. This clinical-grade therapy could provide significant patient benefits.

COMMERCIAL ANGLE

Development of a scalable, off-the-shelf cell therapy for remyelination in Multiple Sclerosis and other demyelinating neurological disorders.

COMMERCIALISATION METRICS

Novelty

6/10

While the use of hESC-derived astrocytes is a sophisticated approach to remyelination, the 'bystander mechanism' and the focus on microglial debris clearance are established concepts in neuroregeneration, making this more of an incremental application of cell therapy than a groundbreaking scientific paradigm shift.

Commercial Potential

8/10

The therapy uses clinical-grade, scalable hESC-derived cells targeting a high-unmet-need market (demyelinating disorders) with demonstrated safety and a clear mechanism of action. The 'clinical-grade' status and scalability suggest it is already positioned for translational development rather than being purely theoretical.

Market Size

9/10

The therapy targets demyelinating disorders (such as Multiple Sclerosis), which represent a massive, high-value patient population with significant unmet medical needs. The mechanism's ability to address multiple facets of remyelination failure suggests a broad, multi-indication platform potential within the neurodegenerative market.

Tech Readiness

5/10

The technology has transitioned from basic lab discovery to 'clinical-grade' manufacturing and has demonstrated efficacy in animal models (lysolecithin-induced), but it remains in the preclinical validation stage rather than undergoing human clinical trials.

IP Strength

8/10

The development of a 'clinical-grade' hESC-derived astrocyte product (AstroRx) suggests a proprietary, scalable manufacturing process and a specific cell composition that is highly patentable. The multi-modal mechanism of action—addressing debris clearance and OPC differentiation—provides multiple layers of therapeutic defensibility and platform potential for various CNS disorders.

[View in Dashboard](#) [Open Paper](#)

2. sc-rDSeq: a robust and cost-effective full-length total RNA sequencing method **35/50** for single cells reveals multilayered heterogeneity in drug-resistant lung cancer cells.

"Next-gen single-cell RNA sequencing reveals hidden disease drivers for precision medicine."

Main researcher: Oren Ram

Department of Biological Chemistry, Alexander Silberman Institute of Life Sciences, The Hebrew University, 9190401 Jerusalem, Israel.

Published in Nucleic acids research · 2026-04

WHY NOW

Offers superior data resolution at lower cost, poised to become a foundational tool in advanced diagnostics and drug discovery, especially for complex diseases like cancer.

COMMERCIAL ANGLE

The technology presents a high-value platform for next-generation precision diagnostics and drug discovery tools by providing more comprehensive transcriptomic data for identifying rare disease drivers.

COMMERCIALISATION METRICS

Novelty

8/10

The technology represents a significant platform innovation by enabling full-length, non-polyA RNA capture in a scalable droplet format, overcoming a major technical limitation of current industry standards like 10x Genomics. This capability to access the 'dark transcriptome' (ncRNA, histone RNA, etc.) provides a new dimension of biological insight that is highly relevant for precision oncology.

Commercial Potential

8/10

The technology offers a significant, platform-enabling improvement over industry standards (10x Genomics) by capturing non-polyA RNA and providing higher UMI density, which has clear commercial utility for drug discovery and precision oncology. Its scalability and cost-effectiveness position it as a highly licensable tool for large-scale clinical biomarker profiling.

Market Size

8/10

The technology targets the high-value single-cell sequencing market by offering a platform-enabling solution for comprehensive transcriptomics, which is essential for advanced drug discovery and personalized oncology. Its ability to capture non-polyA RNAs and alternative splicing at scale provides a significant competitive edge in the expanding precision medicine and biomarker discovery sectors.

Tech Readiness

4/10

The technology is currently at the proof-of-concept stage, demonstrated through successful validation in a laboratory setting using cancer cell lines. While it shows significant performance improvements over existing methods, it has not yet been transitioned into a standardized, commercially available kit or integrated into clinical diagnostic workflows.

IP Strength

7/10

The technology represents a potentially patentable platform innovation by optimizing droplet-based capture for non-polyA RNA, offering a clear technical advantage over industry standards like 10x Genomics. However, defensibility may be challenged by the crowded landscape of single-cell sequencing chemistries and the difficulty of protecting specific microfluidic or bead-based optimizations once widely adopted.

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3. The effect of a proprietary chlorine-stabilizing surface-coating formulation on healthcare environment disinfection in multidrug-resistant bacteria isolation rooms. 33/50

"Durable antimicrobial coating drastically reduces hospital superbug transmission on surfaces."

Main researcher: Oren Zimhony · Oren_Z@clalit.org.il.

Infectious Diseases Unit, Faculty of Medicine, Kaplan Medical Center, Hebrew University of Jerusalem, Pasternak St 1 Rehovot, Rehovot, POB1 76100, Israel. Oren_Z@clalit.org.il.; Faculty of Medicine, Hebrew University of Jerusalem, Jerusalem, Israel.

Oren_Z@clalit.org.il.

Published in Antimicrobial resistance and infection control · 2026-06

WHY NOW

Provides an immediate, scalable solution to combat rising multidrug-resistant infections, crucial for hospital safety, operational efficiency, and reducing healthcare-associated costs.

COMMERCIAL ANGLE

This technology offers a scalable solution for hospitals and long-term care facilities to enhance infection control protocols and reduce the transmission of multidrug-resistant organisms through high-durability antimicrobial coatings.

COMMERCIALISATION METRICS

Novelty

5/10

The research presents an incremental improvement on existing disinfection protocols by using a proprietary coating to extend the efficacy of standard chlorine-based agents. While clinically useful, the innovation is an additive formulation rather than a groundbreaking paradigm shift in antimicrobial science.

Commercial Potential

8/10

The technology demonstrates a clear, additive value proposition to existing hospital disinfection protocols and uses a validated proof-of-concept to address high-value clinical needs (MDRO reduction). The 'proprietary formulation' suggests protectable IP that can be easily commercialized as a specialized chemical adjunct for the healthcare sector.

Market Size

6/10

The technology addresses a high-value niche in infection control (MDRO isolation rooms), but its application as an adjunct to existing protocols suggests a specialized additive market rather than a broad-scale replacement for global disinfection commodities.

Tech Readiness

7/10

The technology has moved beyond lab-scale validation into a controlled quasi-experimental clinical setting with real-world implementation in hospital isolation rooms. It is currently at the stage of clinical validation/pilot testing, positioned just before full-scale commercial rollout or large-scale multi-center trials.

IP Strength

7/10

The use of a proprietary formulation (OxiLast™) to stabilize existing disinfectants suggests a protectable chemical composition or method-of-use patent. However, the efficacy relies on an additive application to standard protocols, which may face moderate defensibility challenges against generic formulations or easier-to-replicate chemical stabilizers.

[View in Dashboard](#) [Open Paper](#)

4. Multiple Charged Purine (MCP) PNA as a Simple Mode for Cellular Uptake.

33/50

"Novel synthetic DNA analogue delivers therapeutics directly into cells, bypassing complex vectors."

Main researcher: Eylon Yavin

The Institute for Drug Research, The School of Pharmacy, The Faculty of Medicine, The Hebrew University of Jerusalem, Hadassah Ein-Kerem, Jerusalem 9112102, Israel.

Published in ACS polymers Au · 2026-06

WHY NOW

Solves a major delivery challenge in genetic therapies, enabling safer and more efficient intracellular targeting for a wide range of antisense therapeutics and genetic interventions.

COMMERCIAL ANGLE

Development of a versatile, high-stability platform for intracellular genetic targeting and antisense therapeutics without the need for expensive lipid nanoparticles or viral vectors.

COMMERCIALISATION METRICS

Novelty

6/10

The work provides a practical chemical modification to improve PNA uptake, but utilizing charge modification for cellular delivery is a known strategy in oligonucleotide chemistry rather than a groundbreaking platform shift.

Commercial Potential

8/10

This represents a platform-enabling technology that solves a fundamental bottleneck (cellular uptake) for PNA, significantly expanding the addressable market for PNA-based therapeutics and diagnostics.

Market Size

9/10

The work addresses a fundamental delivery bottleneck for PNA, a platform with massive potential across antisense therapy, diagnostics, and gene editing markets. Solving cellular uptake enables a wide range of therapeutic applications, making this a platform-enabling advancement with a very large total addressable market.

Tech Readiness

3/10

The work focuses on fundamental chemical modification of PNA for cellular uptake, representing basic proof-of-concept laboratory research without evidence of in vivo efficacy or a developed delivery platform.

IP Strength

7/10

The introduction of a specific chemical modification (MCP) to solve a fundamental PNA delivery bottleneck is highly patentable as a composition of matter. While it enables a platform for cellular uptake, its ultimate strength depends on the breadth of the claims regarding the purine modifications.

[View in Dashboard](#) [Open Paper](#)

5. Total Synthesis of Conjugation-Ready Sulfated Red Algae Carrageenan Oligosaccharides for Sensing Applications.

31/50

"Precision-engineered seaweed oligosaccharides unlock advanced drug delivery and biosensing."

Main researcher: Mattan Hurevich

Institute of Chemistry and Center for Nanoscience and Nanotechnology, The Hebrew University of Jerusalem, Safra Campus, Givat Ram, Jerusalem 9190401, Israel.

Published in Journal of the American Chemical Society · 2026-06

WHY NOW

Opens new avenues for targeted drug delivery systems and highly sensitive diagnostics by providing controlled, customizable biological building blocks, previously unattainable.

COMMERCIAL ANGLE

The ability to manufacture highly controlled, homogeneous carrageenan building blocks enables the development of advanced targeted drug delivery systems and highly sensitive biosensors.

COMMERCIALISATION METRICS

Novelty

8/10

The work represents a significant leap in glycan engineering by achieving the first total synthesis of homogeneous carrageenan oligogalactans, overcoming long-standing structural complexity barriers. This provides a platform-enabling methodology for precision sulfated polysaccharides that was previously inaccessible via natural extraction.

Commercial Potential

6/10

The paper provides a high-value chemical platform for creating precise, functionalized building blocks, but the commercial application is currently limited to niche sensing and R&D tools rather than a large-scale industrial material.

Market Size

7/10

The ability to produce homogeneous, precisely sulfated carrageenan oligosaccharides unlocks high-value niche markets in precision drug delivery and advanced diagnostic sensing. However, while the platform is enabling, the total addressable market is currently constrained by the specialized, high-margin nature of synthetic glycans compared to bulk commodity carrageenans.

Tech Readiness

3/10

The work is at a fundamental academic stage, focusing on the complex chemical proof-of-concept for total synthesis rather than scalable manufacturing or functional application testing. While the 'conjugation-ready' nature provides a pathway for future platforms, the technology remains in the basic research/lab-scale phase.

IP Strength

7/10

The development of a specialized, multi-level protecting group hierarchy for total synthesis of complex polysaccharides represents a high barrier to entry and strong potential for composition-of-matter patents. While the synthetic methodology is highly defensible, its commercial impact depends on the successful application of these oligosaccharides in specific high-value sensing or therapeutic markets.

[View in Dashboard](#) [Open Paper](#)

6. Phosphoproteomic profiling reveals reversal of dysregulated kinase signaling **28/50** by nitric oxide inhibition in the Shank3 mouse model of autism.

"Targeting nitric oxide reveals new precision pathways for autism spectrum disorder therapeutics."

Main researcher: Haitham Amal · Haitham.amal@mail.huji.ac.il.

Institute for Drug Research, School of Pharmacy, Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel. Haitham.amal@mail.huji.ac.il; The Rosamund Stone Zander and Hansjoerg Wyss Translational Neuroscience Center, Boston Children's Hospital, Boston, MA, USA. Haitham.amal@mail.huji.ac.il.

Published in Scientific reports · 2026-06

WHY NOW

Identifies specific, actionable molecular targets for developing much-needed precision pharmacological interventions, addressing underlying mechanisms in autism spectrum disorders.

COMMERCIAL ANGLE

The findings provide high-resolution molecular targets for the development of precision pharmacological interventions aimed at correcting dysregulated kinase signaling in autism spectrum disorders.

COMMERCIALISATION METRICS

Novelty

6/10

The study provides meaningful mechanistic insight by mapping the signaling pathways downstream of NO inhibition, but it follows a predictable biological logic rather than introducing a groundbreaking new paradigm or platform technology.

Commercial Potential

6/10

The study identifies a mechanistic target (nNOS inhibition) for a high-unmet-need indication, but the phosphoproteomic focus is primarily descriptive rather than proposing a novel, druggable modality or proprietary platform.

Market Size

9/10

Autism spectrum disorder represents a massive, global therapeutic market with significant unmet medical need and a high prevalence across diverse demographics. Targeting a fundamental neurobiological pathway like nNOS/phosphoproteomics offers potential for high-value, broad-spectrum clinical application.

Tech Readiness

3/10

The research is currently at the fundamental discovery stage, utilizing animal models (Shank3 mice) to identify mechanistic signaling pathways rather than testing a clinical candidate in humans. While it provides high-value biological validation, it lacks the preclinical development or pharmacological optimization required for higher TRL scores.

IP Strength

4/10

The paper describes fundamental mechanism-of-action (MoA) discovery rather than a novel composition of matter; while identifying signaling pathways is valuable for target validation, the use of nitric oxide inhibition is an established biological phenomenon that offers limited new patentable space.

[View in Dashboard](#) [Open Paper](#)

7. Universal rhythmic architecture uncovers two modes of neural dynamics.

28/50

"Universal brain activity patterns enable next-gen personalized neurodiagnostics and BCI."

Main researcher: Ayelet N Landau

Department of Psychology, The Hebrew University of Jerusalem, Jerusalem, 9190401, Israel.; Department of Cognitive and Brain Sciences, The Hebrew University of Jerusalem, Jerusalem, 9190401, Israel.; Department of Experimental Psychology, University College London (UCL), London, WC1H 0AP, UK.

Published in Nature communications · 2026-05

WHY NOW

Offers a foundational framework for more accurate brain monitoring, powering breakthroughs in personalized neurological care, psychiatric assessment, and human-machine interfaces.

COMMERCIAL ANGLE

This universal architecture can be used to develop highly personalized neurodiagnostic software and advanced brain-computer interfaces that distinguish between stable cognitive states and rapid sensory responses.

COMMERCIALISATION METRICS

Novelty

8/10

The research shifts the fundamental paradigm of neural analysis from a frequency-only model to a two-dimensional architecture (frequency + rhythmicity), providing a potentially transformative framework for neurotechnological diagnostics. This provides a platform-enabling discovery that unifies disparate datasets across species and clinical states into a single organizing principle.

Commercial Potential

5/10

While the framework offers high potential for improving the precision of neurodiagnostic biomarkers and personalized EEG analysis, it remains a fundamental descriptive discovery rather than a specific device or therapeutic intervention. Its commercial value depends entirely on its integration into future clinical software or neurotechnology platforms.

Market Size

8/10

The findings provide a platform-enabling framework for universal neurophysiological analysis, addressing a massive market spanning clinical diagnostics, neuroprosthetics, and brain-computer interfaces across diverse patient populations.

Tech Readiness

3/10

The paper presents fundamental discovery-based neuroscience that provides a new theoretical framework for understanding neural dynamics. While it offers potential for future diagnostic biomarkers, it is currently at a basic research stage without a defined prototype, hardware integration, or clinical validation pipeline.

IP Strength

4/10

The work describes a scientific framework and a descriptive method for analyzing existing neural data rather than a novel hardware device or a proprietary biochemical entity. While the 'rhythmicity measure' could potentially be patented as a diagnostic algorithm, the primary output is foundational knowledge that is difficult to defend against competitors using alternative mathematical modeling.

[View in Dashboard Open Paper](#)

8. Identifying maximally informative signal-aware representations of single-cell data using the information bottleneck. 28/50

"AI distills complex single-cell data, pinpointing disease biomarkers for drug discovery."

Main researcher: Mor Nitzan · mor.nitzan@mail.huji.ac.il.

School of Computer Science and Engineering, The Hebrew University of Jerusalem, Jerusalem, Israel; Racah Institute of Physics, The Hebrew University of Jerusalem, Jerusalem, Israel; Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel.

Electronic address: mor.nitzan@mail.huji.ac.il.

Published in Cell systems · 2026-05

WHY NOW

Addresses the data overload in single-cell genomics, providing a powerful, high-throughput software platform to accelerate precision medicine and identify novel disease drivers.

COMMERCIAL ANGLE

The framework can be commercialized as a high-throughput software platform for drug discovery and precision medicine to identify specific cellular biomarkers and disease drivers from noisy single-cell datasets.

COMMERCIALISATION METRICS

Novelty

6/10

The paper applies an established information theory framework (Information Bottleneck) to a well-trodden domain (scRNA-seq), representing an incremental methodological improvement rather than a fundamental scientific breakthrough. While the application to gene clustering and hierarchical structure is useful, it follows existing trends in deep learning-based dimensionality reduction and feature selection.

Commercial Potential

6/10

The framework presents a strong platform-enabling computational tool for drug discovery and biomarker identification, but its commercial value is currently tied to software/algorithm licensing rather than a standalone proprietary product. Its utility depends on integration into existing bioinformatics pipelines or single-cell analysis platforms.

Market Size

8/10

The framework acts as a platform-enabling technology for the massive and growing single-cell multi-omics market, offering high value for drug discovery and precision medicine. By optimizing signal extraction in complex datasets, it addresses a critical bottleneck in the multi-billion dollar bioinformatics and clinical diagnostics sectors.

Tech Readiness

4/10

The paper presents a computational framework and algorithmic methodology (bioIB) rather than a physical product or clinical tool. While it demonstrates utility on existing datasets, it remains at the fundamental software/algorithm development stage (TRL 3-4) rather than a validated industrial platform.

IP Strength

4/10

The innovation is primarily a computational framework based on the Information Bottleneck method, which is an established theoretical principle in machine learning, making the core algorithm difficult to patent. While the application to scRNA-seq is valuable, the defensibility is low as it likely falls under 'mathematical methods' and is susceptible to rapid replication by open-source community developments.

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9. RadD from *Fusobacterium nucleatum* engages NKp46 to promote antitumor cytotoxicity.

28/50

"Bacterial protein activates NK cells, sparking new immunotherapy approaches for cancer."

Main researcher: Ofer Mandelboim

The Concern Foundation Laboratories at the Lautenberg Center for Immunology and Cancer Research, Institute for Medical Research Israel Canada (IMRIC), Hebrew University Hadassah Medical School, Jerusalem, Israel.

Published · 2026-05

WHY NOW

Uncovers a novel mechanism to enhance natural killer cell activity, offering a promising avenue for developing new, potent cancer immunotherapies with potentially broader applications.

COMMERCIAL ANGLE

Development of RadD-based NK cell agonists or biomimetic immunotherapies to enhance natural killer cell cytotoxicity in oncology.

COMMERCIALISATION METRICS

Novelty

7/10

The discovery of a specific bacterial protein (RadD) acting as a direct ligand for the NKp46 receptor is highly novel and expands the known ligandome of NK cells. While it provides a strong foundation for targeted immunotherapies, it is an incremental discovery of a molecular interaction rather than a platform-level technological breakthrough.

Commercial Potential

6/10

The identification of RadD as a specific ligand for NKp46 provides a clear molecular target for developing NK cell agonists or mimics to enhance antitumor cytotoxicity. However, the commercial path is complicated by the dual nature of *F. nucleatum* in cancer progression and the need for precise delivery to avoid systemic inflammation.

Market Size

6/10

The target market encompasses oncology immunotherapy, specifically head and neck cancers, which is substantial; however, the current focus is on a highly specific host-microbe interaction rather than a broad platform application.

Tech Readiness

3/10

The research is at the fundamental discovery stage, establishing a biological mechanism through in vitro assays and murine models. While it identifies a novel target for immunotherapy, there are no lead candidates, pharmacokinetic data, or clinical validation steps presented.

IP Strength

6/10

The identification of a specific bacterial ligand (RadD) for a known human receptor (NKp46) provides a clear target for composition-of-matter patents on RadD-mimetics or agonists. However, since RadD is a naturally occurring bacterial protein, the IP strength depends on the ability to engineer novel derivatives or specific delivery vehicles rather than the discovery of the axis itself.

[View in Dashboard](#) [Open Paper](#)

10. Nitric oxide inhibition ameliorates cortical proteomic changes in the Cntnap2 27/50

"Nitric oxide inhibition reverses molecular deficits in genetic autism models, guiding new therapies."

Main researcher: Haitham Amal · Haitham.amal@mail.huji.ac.il.

Institute for Drug Research, School of Pharmacy, Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel.
Haitham.amal@mail.huji.ac.il; Rosamund Stone Zander and Hansjoerg Wyss Translational Neuroscience Center, Boston Children's Hospital, Harvard Medical School, Harvard University, Boston, MA, USA. Haitham.amal@mail.huji.ac.il.
Published in Molecular autism · 2026-04

WHY NOW

Provides a specific therapeutic strategy by addressing underlying molecular pathology in genetically defined autism subtypes, a high-need area with limited targeted treatments.

COMMERCIAL ANGLE

The findings support the development of small-molecule therapeutics targeting nitric oxide synthase as a precision medicine approach for genetically defined autism subtypes.

COMMERCIALISATION METRICS

Novelty

5/10

The study follows a logical progression from previously established findings (implicated NO signaling) to mechanistically characterizing those findings via proteomics. While it provides important biological validation, the innovation is incremental rather than a groundbreaking discovery of a new pathway or a novel therapeutic modality.

Commercial Potential

6/10

The study identifies a specific molecular target (nNOS inhibition) and a drug (7-NI) that shows functional efficacy in two different genetic models of ASD, providing a clear path toward drug repurposing or development. However, the commercial potential is limited by the complexity of treating neurodevelopmental disorders and the significant translational gap between mouse proteomic correction and human clinical efficacy.

Market Size

9/10

The research targets Autism Spectrum Disorder (ASD), a massive global market with high unmet medical need and a large patient population. Furthermore, the identification of a specific, conserved molecular mechanism (NO signaling) suggests a potential platform for treating multiple genetic subtypes of ASD.

Tech Readiness

3/10

The research is in the early preclinical stage, focusing on mechanistic proteomic profiling and behavioral validation in murine models. While it identifies a potential therapeutic target (nNOS inhibition), it lacks the pharmacokinetic, safety, or dose-escalation data required for clinical translation.

IP Strength

4/10

The innovation relies on 7-Nitroindazole, a well-known, off-patent small molecule, which limits the ability to claim novel composition of matter. While the proteomic targets identified could potentially support new method-of-use patents, the underlying mechanism is incremental rather than a breakthrough platform technology.

[View in Dashboard](#) [Open Paper](#)

11. Model-Based Attribution of Biologic Effectiveness in Psoriasis: Drug vs Context Contributions.

26/50

"AI predicts biologic drug effectiveness, revolutionizing personalized psoriasis treatment."

Main researcher: Yuval Ramot · yuval.ramot@mail.huji.ac.il.

Faculty of Medicine, Hebrew University of Jerusalem, Jerusalem, Israel; Service for Inflammatory Diseases of the Skin, Department of Dermatology, Hadassah Medical Center, Jerusalem, Israel. Electronic address: yuval.ramot@mail.huji.ac.il.

Published in Journal of the American Academy of Dermatology · 2026 Jul 15

WHY NOW

Reduces trial-and-error prescribing in immunology, saving healthcare costs and improving patient outcomes through data-driven precision medicine in chronic autoimmune conditions.

COMMERCIAL ANGLE

Development of a precision medicine software tool for predicting patient response to biologic therapies, reducing trial-and-error prescribing in immunology.

COMMERCIALISATION METRICS

Novelty

6/10

The approach of using model-based attribution to decouple drug effects from patient context is a sophisticated application of causal inference/ML, but it builds on existing attribution frameworks rather than inventing a new scientific paradigm.

Commercial Potential

6/10

The work suggests a high-value application in precision medicine and companion diagnostics for biologics, but without the abstract, it is unclear if this is a general theoretical framework or a scalable, proprietary tool.

Market Size

8/10

Psoriasis is a high-prevalence chronic condition with a massive, high-value biologics market; tools that optimize drug effectiveness and patient stratification represent a significant commercial opportunity.

Tech Readiness

3/10

The title indicates a model-based theoretical or analytical framework for attributing drug effectiveness, which is characteristic of early-stage computational research (TRL 3) rather than a validated clinical tool or product.

IP Strength

3/10

The title suggests a methodological or diagnostic framework for attribution analysis, which is typically difficult to patent and often categorized as a mathematical method or software tool with low inherent defensibility.

[View in Dashboard](#) [Open Paper](#)

12. Strategic Template Filtering Accelerates Fragment-Based Peptide Docking. 26/50

"Accelerated AI peptide docking speeds drug discovery, enabling efficient therapeutic design."

Main researcher: Ora Schueler-Furman

Microbiology and Molecular Genetics & Quantitative Molecular Medicine, The Hebrew University of Jerusalem, Jerusalem 91120, Israel.

Published in Journal of chemical information and modeling · 2026-06

WHY NOW

Significantly cuts computational time and cost, enhancing the efficiency of AI-driven platforms for developing novel peptide-based drugs with high binding affinity and specificity.

COMMERCIAL ANGLE

Licensing or integrating this high-efficiency docking algorithm into AI-driven drug discovery platforms to accelerate the design of peptide-based therapeutics.

COMMERCIALISATION METRICS

Novelty

4/10

The work describes an incremental optimization of an existing protocol (PatchMAN) by adding filtering and localization to reduce compute time. It is an efficiency improvement rather than a groundbreaking shift in methodology or a novel scientific discovery.

Commercial Potential

6/10

The tool provides a clear efficiency gain for an existing pipeline, making peptide-protein docking more accessible for drug discovery. However, it is an incremental optimization of a protocol rather than a novel, standalone platform or a direct therapeutic lead.

Market Size

8/10

Peptide-protein interaction modeling is a foundational platform technology for the multi-billion dollar biologics and drug discovery markets. Reducing the computational cost of high-accuracy docking directly enables larger-scale virtual screening and accelerates the development of peptide-based therapeutics.

Tech Readiness

4/10

The work is a computational optimization of an existing software protocol, moving it from a high-cost academic tool to a more efficient one, but it remains in the 'tool development' stage without evidence of integrated drug discovery pipeline application.

IP Strength

4/10

The innovation is an incremental optimization of an existing open-source protocol (PatchMAN) focused on computational efficiency rather than a novel, proprietary algorithm. As a software refinement for fragment filtering, it is likely viewed as an obvious improvement in the field, making it difficult to secure strong, broad patents.

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